

## When Treatment Ends but Illness Persists: How Should Medicine Define and Treat Persistent Symptoms After Lyme Disease?

A Creative Dialogue Between Meghan O'Rourke and Allen Steere

### Setting

Years after their first encounter, O'Rourke has come to accept that improvement after antibiotics cannot by itself prove persistent infection, although she still believes it deserves serious clinical attention. Steere still opposes prolonged antibiotic treatment without evidence of ongoing infection, but he now recognizes that uncertainty should never become a reason to dismiss suffering or withdraw care.

O'Rourke and Steere had just concluded a medical humanities seminar on Lyme disease. The other students had already left, and a joint statement remained on the screen:

Patients who believe they have chronic Lyme disease often continue to report nonspecific symptoms even after receiving adequate antibiotic treatment. Although these symptoms can be distressing, the current evidence does not support long-term antibiotic use.

At the bottom of the screen read: "Please co-edit and sign this together."

### Part 1

Steere politely deferred for a moment, but without waiting for consultation, he rushed to the computer and began editing the text.

O'Rourke: Did you see that? That's part of the problem. We just read Thornber's critique that medicine sometimes involves "speaking for patients instead of speaking with people."

Steere: Do you think we need to each take one hand and cover half the keyboard to qualify as person-focused care?

O'Rourke: It's not that extreme, but what I'm saying is that revising this joint statement requires your professional expertise and my lived experience.

At that moment, O'Rourke pushed Steere's hand aside and deleted the words "believe they have."

Steere: Why delete that? You know "chronic Lyme disease" is controversial. I still believe the standard course of antibiotics has already killed the bacteria in the body, and the persistent symptoms are more likely caused by an autoimmune response.

O'Rourke: But the word "believe" makes readers subjectively doubt patients' lived experiences from the start. The fatigue, joint pain, and cognitive impairment we actually experience aren't just a belief.

Steere: I'm not denying the reality of the symptoms, but I cannot agree that you patients' interpretation of what causes them.

O'Rourke: But you should know that you were one of the first to define Lyme disease. If you write now that patients merely "believe" they have chronic Lyme disease, it will only hurt more patients.

Steere (after careful consideration): Well, I guess I'll delete it then. But we must also make a clear distinction between symptoms and a proven persistent infection.

## Part 2

O'Rourke: Speaking of your experience identifying Lyme disease, the fact is that you did so precisely because you took "lived experience" and patients' observations seriously. In 1975, children in Old Lyme were thought to have juvenile rheumatoid arthritis. It was several mothers who insisted that these cases were too concentrated to be a coincidence.

Steere: They provided crucial clues. We noticed in the medical records that the children's rashes coincided with the tick-active season.

O'Rourke: You initially considered a viral cause.

Steere: Because the pathogen hadn't been identified yet. Later, Willy Burgdorfer discovered spirochetes in ticks, and research confirmed that this disease in the United States is primarily caused by *Borrelia burgdorferi*. As the evidence changed, so did our explanation.

O'Rourke: This ties in perfectly with the "tolerance of ambiguity" we just discussed in Bleakley's articles. Both in the past and today, the medical community needs to acknowledge that the available evidence may be incomplete, and that symptoms persisting after antibiotic treatment may be part of chronic Lyme disease.

Steere: But tolerating uncertainty does not mean treating every explanation as equally reliable. The medical community's perspective is supported by data from human clinical trials, whereas your evidence actually falls more within the realm of clinical observation. Furthermore, the early history you mentioned does not prove that every case of unexplained fatigue and pain is Lyme disease. Early-stage Lyme disease has relatively clear evidence of infection, such as tick exposure, an expanding erythema migrans, and subsequent joint, neurological, or cardiac problems. The current controversy primarily centers on the period following treatment.

O'Rourke: So you don't deny the existence of Lyme disease, nor do you oppose the use of antibiotics to treat patients with clear evidence of infection.

Steere: Certainly not at this point. Rather than debating whether symptoms persisting after the completion of an antibiotic course are part of chronic Lyme disease, the real issue now is this: if symptoms persist after the recommended course of antibiotics, can this prove that the infection is still present? And in the absence of evidence of ongoing infection, should antibiotic treatment continue for months or even years?

Steere: If a patient originally had clear evidence of Lyme disease, the CDC and IDSA typically refer to this as "post-treatment Lyme disease syndrome," or PTLDS. This term acknowledges the persistence of symptoms without presuming that live bacteria are still causing an ongoing infection.

O'Rourke: The International Lyme and Associated Diseases Society and our patients use the term "chronic Lyme disease" because they believe the bacteria may remain in deep tissues and were not completely eradicated by short-term treatment, so they advocate for longer courses of antibiotics.

Steere: Moreover, "chronic Lyme disease" is sometimes applied to people who have never had clear evidence of infection. As a result, several distinct patient groups are lumped under the same diagnosis, and other potential causes may be overlooked.

O'Rourke: I don't believe that every long-term symptom proves a persistent infection, nor do I believe that everyone should take antibiotics indefinitely.

Steere (nodding): On this point, you differ from some of the most radical advocates.

O'Rourke: But for many patients, "chronic Lyme disease" is the first label that explains why they suddenly can't work, read, or care for their children.

Steere: Medicine cannot treat an unproven cause as fact simply to give a name to an experience.

O'Rourke: But you also can't dismiss that experience as if nothing happened just because the cause hasn't been proven.

### Part 3

Steere moved the cursor to "nonspecific symptoms."

Steere: This term needs to be highlighted. Muscle fatigue, joint pain, and brain fog can stem from infections, autoimmune diseases, sleep disorders, medication side effects, or depression—these symptoms alone cannot pinpoint a single cause.

O'Rourke: But "nonspecific" explains that the symptoms don't point to a single cause; it doesn't address how severe the symptoms are.

Steere: I didn't say they were mild.

O'Rourke: But from the patient's perspective, this becomes a cold, clinical term, lacking any human warmth. It's as if everyone gets tired, everyone forgets their words—it's perfectly normal. We just read about Thornber's distinction between "disease" and "illness." "Disease" refers to what appears in test results, pathology reports, and diagnostic labels, while "illness" is how a person lives with it day to day. Your text accurately describes the former but says almost nothing about the latter.

Steere (thinking, somewhat pained): Then give me an example—how would you say one truly touches upon "illness"?

O'Rourke: I'm standing at the front of a poetry class, and suddenly I can't remember the word "spring"; all I can say is "the season after winter, when flowers start to grow." I look at someone I've known for years, yet for a moment I don't know who he is. Going out for dinner might mean spending the next three days in bed to make up for it.

O'Rourke: When Tolstoy wrote about Ivan Ilyich visiting the doctor, the doctor kept trying to determine whether it was wandering kidney, chronic catarrh, or appendicitis. But, "To Ivan Ilyich, only one question was important: was his case serious or not?" In the end, the doctor never answered what Ivan was truly afraid of: Will I be able to continue living my old life?

Steere: A doctor needs to identify the disease first in order to know how to treat it.

O'Rourke: In your subconscious, these two issues have a natural order: first resolve yours, then it's my turn.

O'Rourke: When Nobel wrote about people who had lost some of their ability to speak, those around them began to talk around them, as if they were no longer present—yet "Inside, they are still the same person." Deep down, they remain the same person. Chronic illness can make people "disappear" in this way as well: you can still speak, yet no one believes the changes you describe. This is societal loneliness. It's not that there's no one around you, but that others no longer recognize the person you were before the illness, nor do they acknowledge what you've lost now. The "recognition" a patient needs isn't the doctor's agreement with their diagnosis, but rather the doctor's acknowledgment that, "I can see that your life has indeed changed."

Steere: This can be done even before the cause is determined.

They revised the original sentence to:

These symptoms may include severe fatigue, pain, cognitive difficulties, and impaired daily functioning; the fact that a single cause cannot currently be identified does not mean the symptoms are minor or unreal.

#### Part 4

O'Rourke: During our first conversation, you acknowledged that the antibiotic treatment helped me, but you didn't think that proved a persistent infection.

Steere: I still believe that.

O'Rourke: Before treatment, I would sweat through the night and wake up with my clothes completely soaked; I always felt like I couldn't get enough air; and the pain and brain fog made it increasingly impossible for me to work. After starting treatment, my symptoms actually worsened for a while. Then one day, I woke up and ran three miles. The night sweats stopped, my breathing became easier, and my energy returned.

Steere: Your improvement is real. But at the time, in addition to doxycycline, you were also taking Mepron to treat babesiosis. I remain firmly convinced that your symptoms could have been influenced by other infections, immune responses, autonomic nervous system function, or the combination of medications. That is medical uncertainty. It would be dishonest to simply choose the explanation that best aligns with the patient's wishes.

O'Rourke: But dismissing an explanation that comes from the patient community is not neutral either.

Steere (shaking his head): What I mean is: your treatment response is worth documenting, but it cannot, on its own, answer the question of what caused the illness—whether it was still-active bacteria, a post-treatment immune response, another infection, or a combination of factors. O'Rourke (compromising): Then please state both points; don't just leave it at "it doesn't prove chronic Lyme disease."

Steere: Sure. Improvement is important evidence, but it proves very little.

O'Rourke: That's the full sentence I want patients to hear.

#### Part 5

O'Rourke: I still remember when you used to think I needed to let go of my "Lyme ideology."

Steere: I meant that you are unable to consider evidence that contradicts your beliefs.

O'Rourke: But the word "ideology" actually implies "why is this patient so stubborn?"—it casts patients as irrational. That's stigmatization of Lyme disease patients. Moreover, you scientists can usually retain your credibility even after proposing a false hypothesis. However, if a patient offers a single incorrect explanation, even their descriptions of pain and fatigue may lose credibility as a result.

Steere: But your patient advocacy groups shouldn't call people who disagree with the theory of persistent infection liars, conspirators, or murderers either. I've been cornered at conferences before, approached by people posing as journalists, and I've even received death threats.

O'Rourke: Those actions shouldn't happen. It's true that patients have been neglected in the past, but that doesn't justify harassment. But you have to understand that when patients are suspected by the medical community for a long time, what doctors call "insufficient evidence" turns into "I won't help you anymore."

Steere: But that doesn't mean both sides have equal power.

O'Rourke: Right. In the exam room, it's often still a doctor-down-to-the-patient relationship: You've been involved in the discovery of Lyme disease and published papers. Your judgments can make their way into guidelines, medical records, and insurance policies, but patients can only struggle to prove they aren't exaggerating their symptoms.

Steere: My work influences other doctors, but my opposition to long-term antibiotic use isn't just about professional standing. IV catheter infections, *Clostridioides difficile* infection, and drug side effects can all genuinely harm patients. Even with the LYMErix vaccine, which I led in developing, I took responsibility by specifically outlining potential side effects in *Science*—a move that indirectly led to a sharp drop in vaccine sales.

Steere (adding): Raising a possibility does not equate to proving a causal relationship.

O'Rourke: That is precisely the standard I hope you'll apply to patients' experiences: neither rashly presenting possibilities as conclusions, nor ignoring them simply because they lack conclusive causal evidence.

Steere: That's a fair point.

## Part 6

O'Rourke: I'd also like to ask you this: Suppose we go back to 2014. I come to you with years of symptoms and conflicting test results. Would you prescribe doxycycline for me?

Steere: Based on the evidence available to you at the time, no.

O'Rourke: What about just trying it once?

Steere: Even then, there would need to be sufficient justification. Because a single trial could cause side effects, lead to a misdiagnosis, and increase the risks.

O'Rourke: But when you weigh the risks of treatment, don't you also consider the risk that the patient might lose their ability to work while waiting?

Steere: Yes, the cost of waiting should also be taken into account.

O'Rourke: Exactly—that's what patient autonomy is all about. It's not about the patient asking for a prescription and the doctor simply writing it, nor is it about the doctor saying "no" and the patient walking away; rather, it's about both parties engaging in equal negotiation and communication.

Steere: We should decide together. Only then can we clearly explain the consequences of treatment, the cost of waiting, and alternative approaches to symptom relief.

O'Rourke: In Gawande's *Being Mortal*, Joseph Lazaroff chose a high-risk surgery. The doctors told him the surgery might fail, but they didn't fully help him understand that even if the surgery were technically successful, he might not be able to return to the life he wanted.

Steere: So to truly achieve informed consent, you must first ask the patient what kind of life they most want to preserve.

Steere: What did you most want to preserve at that time?

O'Rourke: Not having to fear that the next word would suddenly vanish while I was teaching; not having to spend three days in bed after walking just a short distance. I wanted to continue being a partner, a mother, and a writer—rather than waking up every day and immediately having to examine my own body.

Steere: All of these can become treatment goals, even if a definitive diagnosis hasn't been established yet.

O'Rourke: That's what "person-focused care" is all about. Doctors don't have to wait until a diagnosis is completely confirmed before they start helping a person get their life back.

Part 7

Steere reorganized the statements on the screen.

Some patients with clear evidence of Lyme disease who have completed the recommended treatment may still experience severe fatigue, pain, cognitive difficulties, and impaired daily functioning afterward. In research and clinical practice, no patient's symptoms should be dismissed simply because the cause is unknown.

Improvements following treatment should be carefully documented and studied. At the same time, improvement alone does not prove the presence of a persistent infection. In the absence of objective evidence of infection or treatment failure, current evidence does not support additional antibiotic therapy. Physicians and patients should all discuss the potential harms of long-term treatment, as well as the potential costs of continuing to wait, while continuing to investigate other possible causes and providing support to alleviate symptoms and restore function. We acknowledge that diagnostic disagreements should not be a reason to discontinue care. In the end, it's all about people. Do not leave the patient alone.

Steere and O'Rourke